Grassmannian, Flag, and Schubert Varieties

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1 Grassmannians, Schubert cells & Schubert varieties

We work over $\mathbb{C}$.

Fix $G = \mathrm{GL}_n(\mathbb{C})$, the group of invertible $n \times n$ matrices.

Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{C}^n$.

Fix $0 < d < n$. The Grassmannian $\mathrm{Gr}(d, n)$ is the set of all d-dimensional linear subspaces $Z \subset \mathbb{C}^n$. Write $X := \mathrm{Gr}(d, n)$ when discussing its Schubert cells and Schubert varieties.

Ex. $\mathrm{Gr}(1, n)$: the 1-dimensional linear subspaces of $\mathbb{C}^n$. Each nonzero vector $v \in \mathbb{C}^n$ spans a line $\langle v \rangle = \{\lambda v : \lambda \in \mathbb{C}\}$; scalar multiples give the same line. $\mathrm{Gr}(1, n) \cong \mathbb{P}^{n-1}$.

Let $B \subset G$ be the Borel subgroup of upper triangular matrices.

$$B = \left\{ \begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & * \end{pmatrix} \,\middle|\, \text{diagonal entries are nonzero} \right\}.$$

For each d-tuple $I = (i_1 < i_2 < \cdots < i_d)$, let $Z_I := \langle e_{i_1}, \ldots, e_{i_d} \rangle \subset \mathbb{C}^n$ be the corresponding coordinate plane.

The Schubert cell indexed by I is the B-orbit $C_I := B \cdot Z_I \subset X$.

The Schubert variety X_I is the Zariski closure of the Schubert cell, $X_I = \overline{C_I} \subset X$.

1.1 Standard flag

Let $F_j = \langle e_1, \ldots, e_j \rangle \subset \mathbb{C}^n$ for $j = 1, \ldots, n$.

The standard flag is the chain $0 \subset F_1 \subset \cdots \subset F_n = \mathbb{C}^n$.

For $\mathrm{Gr}(1, n)$:

Fix the standard flag $F_j = \langle e_1, \ldots, e_j \rangle \subset \mathbb{C}^n$.

For $d = 1$, the Schubert variety $X_i = \{\text{lines } E \subset \mathbb{C}^n : E \subset F_i\}$.

Projectively, $X_i \cong \mathbb{P}(F_i) \cong \mathbb{P}^{i-1}$.

List: $X_1 \cong \mathbb{P}^0 \subset X_2 \cong \mathbb{P}^1 \subset \cdots \subset X_n \cong \mathbb{P}^{n-1}$, a chain of linear subspaces inside $\mathbb{P}^{n-1}$.

1.2 Geometric characterization of Schubert cells / varieties

Now we give a geometric characterization of Schubert cells / varieties.

(i) C_I is the set of all d-dimensional subspaces $Z \subset \mathbb{C}^n$ s.t. for each $j = 1, \ldots, n$, $\dim(Z \cap F_j) = \#\{k : 1 \le k \le d,\ i_k \le j\}$.
(ii) X_I is the set of all d-dimensional subspaces $Z \subset \mathbb{C}^n$ s.t. for each $j = 1, \ldots, n$, $\dim(Z \cap F_j) \ge \#\{k : 1 \le k \le d,\ i_k \le j\}$. Equivalently, $X_I = \bigcup_{J \le I} C_J$, where $J = (j_1 < \cdots < j_d) \le I$ means $j_k \le i_k$ for all k.

$\dim(Z \cap F_j)$ measures how much of Z lies inside the first j coordinate directions.

$\dim(Z \cap F_j)$:

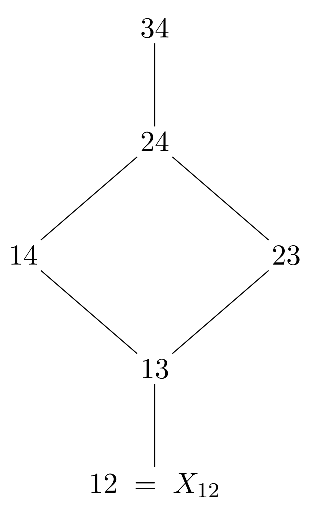
- it gives a nondecreasing sequence $0 \le \dim(Z \cap F_1) \le \cdots \le \dim(Z \cap F_n) = d$.
- For the coordinate plane $Z_I = \langle e_{i_1}, \ldots, e_{i_d} \rangle$, this becomes the number of basis vectors e_{i_k} whose index $i_k \le j$.

So C_I consists of all Z whose intersection dimensions with the flag are the same as for Z_I;

X_I consists of those with dimensions "at least as large."

1.3 Example: $\mathrm{Gr}(2,4)$

Ex. $d = 2, \quad n = 4.$ $\qquad X = \mathrm{Gr}(2,4) = X_{34}$, the 2-planes in $\mathbb{C}^4$.



12	<	13			
13	<	14	13	<	23
14	<	24	23	<	24
24	<	34			

Schubert varieties are indexed by 2-element subsets $I = \{i_1, i_2\} \subset \{1, 2, 3, 4\}$ with $i_1 < i_2$.

The <u>closure order</u> on Schubert varieties is $X_J \subseteq X_I \iff J \le I$, where $J = (j_1 < j_2) \le I = (i_1 < i_2)$ means $j_1 \le i_1$ and $j_2 \le i_2$.

So 12 smallest — can't decrease any further
34 largest — can't increase any further

2 Partial & complete flags

Fix positive integers $d_1, \dots, d_m$ with $d_1 + \cdots + d_m = n$. Here, type means quotient dimensions.

A flag of type $(d_1, \dots, d_m)$ in $\mathbb{C}^n$ is a chain of subspaces $0 = V_0 \subset V_1 \subset \cdots \subset V_m = \mathbb{C}^n$ s.t. $\dim(V_j/V_{j-1}) = d_j$ for $j = 1, \dots, m$.

$\Rightarrow \dim V_j = d_1 + \cdots + d_j$.

The partial flag variety $X(d_1, \dots, d_m)$ is the set of all flags of type $(d_1, \dots, d_m)$ in $\mathbb{C}^n$.

If $m = 2$, take type $(d, n-d)$: a flag is $0 \subset V_1 \subset V_2 = \mathbb{C}^n$ with $\dim V_1 = d$ and $\dim(V_2/V_1) = n - d$.

Since $\dim V_2 = n$, the only free choice is the d-dimensional subspace V_1. Thus $X(d, n-d) \cong \mathrm{Gr}(d, n)$.

$\Rightarrow$ Grassmannians are 2-step partial flag varieties in this convention.

If we take all steps of size 1, $(d_1, \dots, d_n) = (1, \dots, 1)$, then a type $(1, 1, \dots, 1)$ flag is a chain $0 \subset V_1 \subset \cdots \subset V_n = \mathbb{C}^n$ with $\dim V_j = j$.

The full flag variety is $\mathrm{Fl}(\mathbb{C}^n) := X(1, 1, \dots, 1)$, the set of all complete chains of nested subspaces.

A line $\subset$ a 2-plane $\subset \cdots \subset \mathbb{C}^n$.

For $n = 3$, a point of $\mathrm{Fl}(\mathbb{C}^3)$ is a pair $\Big(\underbrace{L}_{\text{line in } \mathbb{C}^3} , \overbrace{P}^{\text{plane in } \mathbb{C}^3} \Big)$ with $L \subset P$.

3 Coordinate flags / Standard flags

Fix the standard basis $e_1, \dots, e_n$ of $\mathbb{C}^n$.

① The standard full flag is $F_\bullet : 0 \subset \langle e_1 \rangle \subset \langle e_1, e_2 \rangle \subset \cdots \subset \langle e_1, \dots, e_n \rangle = \mathbb{C}^n$. The j-th subspace $F_j := \langle e_1, \dots, e_j \rangle$ is the coordinate j-dimensional subspace.

② Standard partial flag

For type $(d_1, \dots, d_m)$ with $d_1 + \cdots + d_m = n$, set $V_j^{\mathrm{std}} = \langle e_1, \dots, e_{d_1 + \cdots + d_j} \rangle$. Then $0 = V_0^{\mathrm{std}} \subset \cdots \subset V_m^{\mathrm{std}} = \mathbb{C}^n$ is the standard flag of type $(d_1, \dots, d_m)$.

① $n = 3$, type $(1, 1, 1)$.

The standard basis is $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$.

$F_1 = \langle e_1 \rangle, \quad F_2 = \langle e_1, e_2 \rangle, \quad F_3 = \langle e_1, e_2, e_3 \rangle = \mathbb{C}^3.$

A point of $\mathrm{Fl}(\mathbb{C}^3)$ is any chain $0 \subset L \subset P \subset \mathbb{C}^3$, e.g. $0 \subset \langle e_1 \rangle \subset \langle e_1, e_2 \rangle \subset \mathbb{C}^3$.

② Type $(1,2)$ in $\mathbb{C}^3$.

Here $n = 3, \quad d_1 = 1, \quad d_2 = 2, \quad d_1 + d_2 = 3.$

A flag of type $(1,2)$ is a chain $0 \subset V_1 \subset V_2 = \mathbb{C}^3$ with $\dim V_1 = 1$ and $\dim(V_2/V_1) = 2$.

Since $\dim V_2 = 3$, V_1 is a line in $\mathbb{C}^3$ and V_2 is the whole space $\mathbb{C}^3$.

So the partial flag variety $X(1,2) =$ all lines in $\mathbb{C}^3 = \mathrm{Gr}(1,3) \cong \mathbb{P}^2$.

The standard flag of type $(1,2)$ is $V_1^{\mathrm{std}} = \langle e_1 \rangle$, $V_2^{\mathrm{std}} = \langle e_1, e_2, e_3 \rangle = \mathbb{C}^3$.

References

[1] V. Lakshmibai and J. Brown, *Flag Varieties: An Interplay of Geometry, Combinatorics, and Representation Theory*, 2nd ed., Texts and Readings in Mathematics, vol. 53, Springer, Singapore; Hindustan Book Agency, New Delhi, 2018.